

Smart Study Focus Monitor

IoT Final Project

Muhammad Asmaq Hussain

Project Idea

- Monitor study focus level
- Use ESP32 sensor to simulate focus
- Send data to computer
- Store and analyze the data

How the System Works :

ESP32 Sensor → MQTT → Python → Database → Graph



LED Alert

Technologies Used :

- ESP32 (Simulated hardware - Wokwi)
- Wokwi (simulation)
- MQTT (communication)
- Python (data processing)
- SQLite (database)
- Matplotlib (visualization)

Live Demo

- Start Python receiver
- Start ESP32 simulation
- Move the potentiometer
- Observe focus values in terminal
- LED turns on when focus is low
- Show graph visualization